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Talk2Biomodels: AI agent-based open-source LLM initiative for kinetic biological models

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Abstract

Background Quantitative kinetic models of biological regulatory processes play an important role in understanding disease mechanisms. However, their simulation and analysis require specialized domain expertise.

Results In this study, we present Talk2Biomodels (T2B), an open-source, user-friendly, large language model-based agentic AI platform designed to facilitate access to computational models of biological systems and promote the FAIRification (Findability, Accessibility, Interoperability, and Reusability) principles in systems biology. T2B allows users to interact with and analyse mathematical models of biological systems through conversations in natural language, thereby lowering the barrier to entry for model interpretation and hypothesis-driven exploration. The platform natively supports models encoded in the Systems Biology Markup Language, a widely adopted standard in the computational biology community. T2B is integrated with the BioModels database (https://www.ebi.ac.uk/biomodels/), enabling retrieval, simulation, and analysis of curated systems biology models. We illustrate the platform's capabilities through use cases in precision medicine, infectious disease epidemiology, and the study of emergent network-level properties in cellular systems — demonstrating how both computational experts and domain scientists without formal modelling training can derive actionable insights from complex biological models. Talk2Biomodels is available at https://github.com/VirtualPatientEngine/AIAgents4Pharma. Detailed documentation and use cases are available at https://virtualpatientengine.github.io/AIAgents4Pharma/talk2biomodels/intro/.

Conclusions In summary, T2B lowers the barrier for non-experts to engage with and extract insights from computational models of biological systems, while simultaneously providing experts with a streamlined interface for analysing models and overall contributes to the FAIRification of models.

Keywords Agentic AI, Systems biology, Foundational models, Ordinary differential equation models

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Background

Recent developments in generative artificial intelligence (AI), particularly in large language models (LLMs), have demonstrated substantial progress. Noteworthy advances in both algorithmic innovation and data-driven biomedical discovery are showcased by the advent of foundational learning models for protein structure, single cell omics and genomic data across species, such as, AlphaFold, scGPT, and Evo 2, to name a few [2, 4, 10]. While LLMs' role is currently confined to narrow, task-specific applications, efforts are underway to enable open-source LLM-driven agentic workflows in biomedical discovery [15]. Collaborative AI agents, with their task orchestration, reflective learning and sophisticated reasoning capabilities, have the potential to revolutionize biomedical research and patient treatment [6].

Quantitative mathematical models have become a crucial component in enhancing our understanding of biological systems. Besides encapsulating the qualitative knowledge of biological interactions, they serve as analytical tools to test the theoretical understanding of the dynamics within biological systems [21].The most common frameworks in systems biology modeling include kinetic, constraint-based, logic, and agent-based modeling. The present study specifically focuses on ordinary differential equation (ODE) models, a subset of kinetic models, due to their widespread application and significance in the field. ODE models represent the most popular modeling approach, with approximately 1662 models[1] currently deposited in the BioModels database [16] and a multitude of models in scientific literature. These models cover a wide range of domains, as demonstrated by their use cases in precision medicine (e.g., inflammatory bowel disease treatment) [5], COVID-19 pandemic studies [23], and theoretical analysis of dynamic systems [18]. Furthermore, ODEs are a standard tool in pharmaceutical research, particularly in the development of physiologically based pharmacokinetic/pharmacodynamic (PBPK/PD) and quantitative systems pharmacology and toxicology (QSP/QST) models [3, 13].

To interact with mathematical models of biological systems, several popular graphical user interface (GUI)-based ODE modelling frameworks exist. These include proprietary software such as the MATLAB toolbox SimBiology [24] and Berkley Madonna [17], as well as open-source alternatives like COPASI [8], the Open Systems Pharmacology Suite [13] and RunBioSimulations [22]. However, both proprietary and open-source solutions typically require software installation, specialized training, and domain knowledge for effective simulation and analysis of models.

To overcome the barriers of existing modelling frameworks, we propose Talk2Biomodels (T2B), an easily accessible chatbot-based ODE modelling platform built using agentic AI (Fig. 1). The T2B platform supports the import of models encoded in SBML [12] and ensures cross-platform interoperability. T2B also provides access to the BioModels database, which hosts the largest collection of curated quantitative mathematical models [16].

Besides the conventional model analysis functionality, such as simulation, parameter scanning, and steady state analysis, T2B can (i) retrieve and display model annotations using standardized names and identifiers, and (ii) perform retrieval-augmented

[1] Viewed on 24.02.2025.

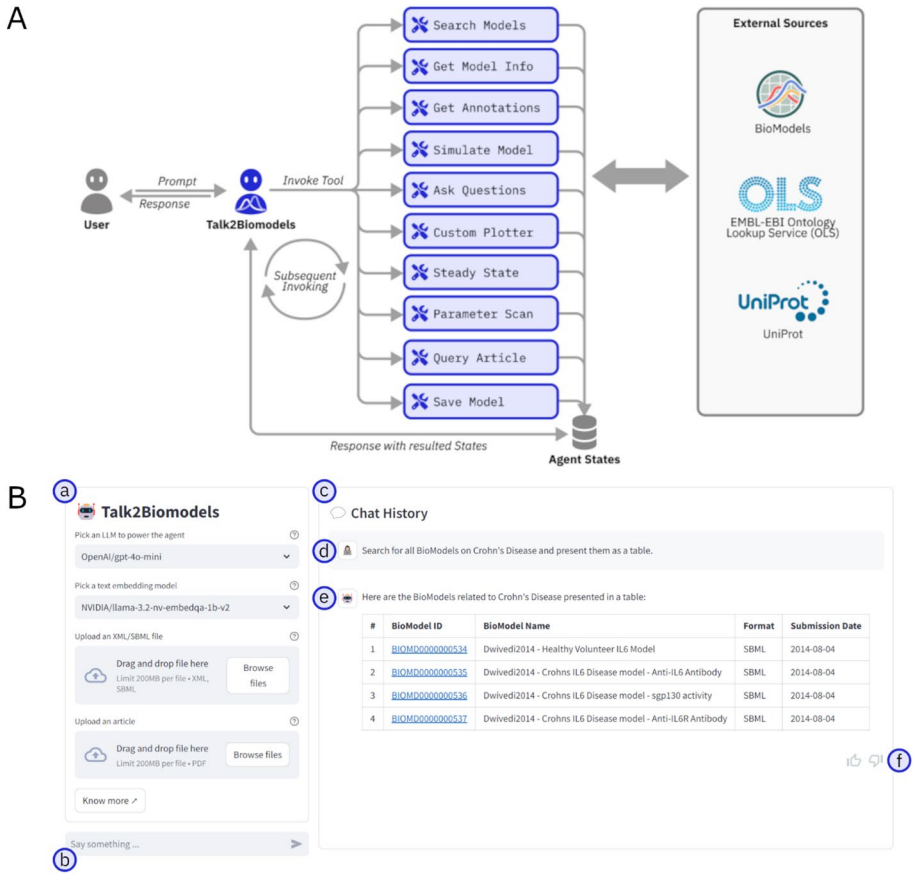

Fig. 1 Architecture diagram and user interface of Talk2Biomodels. **A** Users can prompt the T2B agent to invoke various tools for model search, simulation and plotting, stability analysis, annotation extraction and retrieval-augmented generation (RAG) on uploaded scientific articles. **B** The T2B GUI supports both open and closed-source LLMs, enables users to upload mathematical models in SBML-format and scientific articles as PDFs (a). T2B allows natural language prompts via the chat box interface (b). The GUI features a chat history window (c) displaying question-and-answer conversations (d, e). Additionally, T2B's GUI allows dynamic user feedback through upvote and downvote buttons. (f). Here, we exemplify *search models* tool for keyword-based searches in the BioModels database (e)

generation (RAG) over PDF documents to extract additional insights that inform model interpretation and analysis.

By facilitating the sharing and simulation of mathematical models, T2B contributes to the FAIR principles of data exchange. Furthermore, T2B helps non-experts to interact with ODE models and pose simulation-related questions directly in natural language thus offering an engaging environment for teaching and exploring diversity of models.

Implementation

We developed the T2B application using the LangGraph library (v0.3.2),[2] which provides a framework for constructing stateful computational graphs. In this architecture, each node represents a discrete computational step, while the edges define the relationships between these steps. The application maintains a global state—a snapshot of the application—which is propagated and updated throughout the execution process (Fig. 1).

[2] https://github.com/langchain-ai/langgraph

In the case of T2B, the computational nodes consist of a prebuilt reasoning and action (ReAct)-based agent, implemented following the ReAct paradigm [27], and a suite of tools available to the agent (Fig. 1). Each tool is annotated with metadata describing its capabilities and input/output schema. Upon receiving a user query in natural language, the LLM interprets the intent, selects the appropriate tool(s), and dynamically constructs a sequence of calls to fulfill the request. These agentic AI tools interactively engage with external resources, including BioModels, Ontology Lookup Service (OLS) and UniProt databases [11, 25]. T2B maintains the global state of the agent throughout the chat history. BioModel search, simulations, annotation extraction and stability analysis are carried out using the BASICO library (v0.78) [1]. Directed edges in the graph structure facilitate interactions between the agent and these tools. The agent autonomously selects and executes tool invocations based on user input, dynamically updating the global state during execution. The state management mechanism tracks key-value pairs storing messages, analysis outputs, and intermediate results, ensuring contextual coherence throughout the workflow. Upon computation completion, the last updated state is parsed to extract relevant information and render it for the user.

For configuring agent-based text generation tasks, users can select between two LLMs: OpenAI's GPT-4o-mini [19] and NVIDIA NIM-optimized Meta/Llama-3.3-70B-Instruct.[3] For text-to-embedding transformations, the application supports two embedding models: OpenAI's text-embedding-ada-002 and NVIDIA NIM-optimized Llama-3.2-NV-EmbedQA-1B-V2.[4] The backend of T2B is implemented in Python 3.12, while the front end was developed using Streamlit[5] (v1.39.0) and Plotly[6] (v5.24.1) for an interactive and user-friendly interface.

To evaluate the performance of T2B, we designed a benchmarking strategy utilizing the Task Completion metrics from DeepEval [9], which captures the overall effectiveness of the agent in solving user queries, using LLM-as-a-judge to account for semantic accuracy of responses. To generate evaluation data, we created two synthetic datasets, containing 312 question–answer (Q&A) pairs covering tools *search_models, simulate_model, ask_question, steady_state, custom_plotter and get_modelinfo* (Table S2). Both datasets were designed with textual answers as expected outputs, capturing the heterogeneous nature of tool responses in real-world use cases. For each question, we obtained the agent's ground-truth outputs using BASICO library [1] and assessed performance using the DeepEval library [9], which allowed us to compute the metrics consistently across all datasets.

T2B provides optional integration with LangSmith to enable detailed tracing, debugging, and performance evaluation of agentic workflows. Users can provide feedback in the chat interface through thumbs-up and thumbs-down buttons directly connected to LangSmith, allowing for real-time capture of user satisfaction and facilitating continuous improvement of the agent's performance and reliability. T2B is hosted on GitHub, facilitating open-source contributions and discussions, and is linked to BioModels repository.

[3] https://build.nvidia.com/meta/llama-3_3-70b-instruct/modelcard

[4] https://build.nvidia.com/nvidia/llama-3_2-nv-embedqa-1b-v2

[5] https://streamlit.io/

[6] https://plotly.com/chart-studio-help/citations/

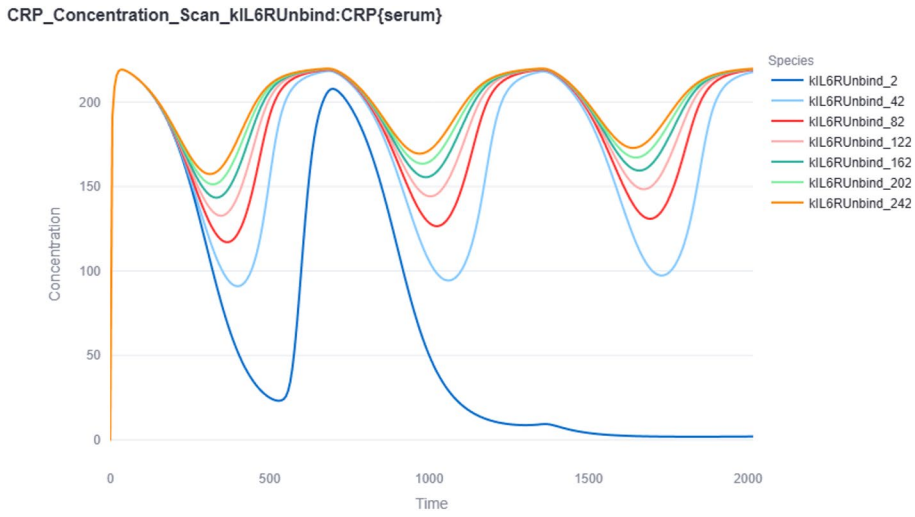

Fig. 2 Parameter scan of a precision medicine model within T2B GUI [5]. A parameter 'kIL6Unbind' is varied in steps of 40 units from 2 to 242 to simulate an unbinding event of an antibody from its target interleukin 6 receptor (IL6R). Each curve represents CRP levels in serum during a separate simulation of this model.

Results and discussion

T2B offers a comprehensive suite of functions for biological and translational ODE models. T2B reduces the risk of inexact responses and hallucinations arising from misinterpreting user-defined simulation requirements through model simulation-specific tool calling and visual representation of the simulation results. Users can upload ODE models in SBML format or retrieve curated models from the BioModels database using the *search models* tool (Fig. 1). The *get model information* tool extracts model details, including species, parameters, and model abstract. Concurrently, the *get annotations* tool retrieves and displays model species' identifiers by dynamically accessing Ontology Lookup Service (OLS) and UniProt data bases [11, 25]. This capability enables researchers to thoroughly explore and interpret the specific nomenclature of the model.

In T2B, SBML models can be directly accessed, simulated, and analyzed. The platform supports time-course simulations with adjustable time steps, initial concentrations and durations (*simulate model* tool), presenting results in dynamic tables and graphs that can be interactively modified (*custom plotter* tool). Additionally, users can interrogate model description and simulation results (*ask questions* tool) and alter model parameters using natural language.

Moreover, T2B offers several tools for model analysis: the *steady state* tool computes whether a model reaches equilibrium, while the *parameter scan* tool performs multiple simulations by systematically varying model initial conditions or parameter values within user-specified range and intervals (Fig. 2). In addition, the *query article* tool leverages retrieval-augmented generation (RAG) framework [14] to extract supplementary information from user-uploaded research articles. Finally, the *save model* tool allows users to save the model they are working with to disk for later modification and reference.

Overall, this suite of agentic AI-driven tools offers a unique platform for working with ODE models, combining user-friendliness and interactive features. In addition, the versatility of agentic-AI framework enables integration of T2B with additional agents, potentially facilitating coordination with tools such as literature search platforms

(Talk2Scholars) and knowledge graphs (Talk2KnowledgeGraphs), thereby enhancing the modeling process and providing modelers with relevant background information.[7]

Use-cases

We present use cases in precision medicine (Table S1), epidemiology, and emergent systems properties of biological networks and make them available on GitHub. With these use cases we demonstrate broad applicability of T2B across different biological domains and how experts and non-experts in computational biology can benefit from T2B. All three examples illustrate the processes of loading, simulating, interactively plotting, and editing models, as well as how LLMs assist in drawing conclusions from the model simulations.

Typically, when a user downloads and simulates a model, they may lack contextual knowledge regarding the simulation duration or the intended objectives of the model. By uploading the associated PDF documentation, such as publication associated with this model, users can enhance their understanding of the simulations of this model. In this context, employing a RAG system on the uploaded PDF assists users in extracting relevant information and identifying key aspects of the model.

Moreover, T2B enables efficient translation of model annotations identifiers to a standard names of the species by communicating with external databases, such as UniProt and EMBL-EBI Ontology Lookup Service over API calls [11, 25]. For instance, *get_annotations* tool could identify the species as a complex or specific isoform (e.g., soluble or not soluble isoform of a cytokine), which could define the scope of the model and have implications in downstream experimental validation tasks.

Precision medicine

This example is based on a published study of a multiscale model of interleukin 6 (IL-6) mediated immune regulation in Crohn's disease to examine drug-disease interaction mechanisms and tocilizumab, an anti-interleukin 6 receptor monoclonal antibody drug, treatment prediction [5].

Here we show how users can simulate and analyze this model via natural language prompting.[8] Within the RAG framework, T2B determines the model's specified simulation time of 2016 h as reported in the publication. In addition, the LLM assists users in identifying parameters related to the drug dosing regimen by querying species names and providing description of model species. Also, T2B optimizes retrieval of model annotations, facilitated drug dose adjustment and streamline model simulation. With T2B users can simulate drug administration at intervals of either 2 or 4 weeks and view the value of the clinical readout parameter (CRP in serum) at any given time point. Additionally, users can determine the time required for the depletion of serum anti-IL6Rα concentration by replicating the published figure [5]. The effect on CRP suppression can also be compared by simulating different antibodies with varying dissociation affinities, thereby performing in silico tests for potential antibodies (Fig. 2).

[7] https://github.com/VirtualPatientEngine/AIAgents4Pharma

[8] Detailed prompts are available at Table S1 and at https://virtualpatientengine.github.io/AIAgents4Pharma/talk2biomodels/cases/Case_1/.

Epidemiology

This example utilizes an epidemiological model to predict the spread of COVID-19[23].[9] T2B enables users to evaluate the effectiveness of quarantine as a COVID-19 containment measure by simulating its impact on the number of infected cases over time, thereby reproducing published findings [23]. Here, specifically, T2B offers a concise representation of model elements and translates parameter names into a user-friendly representation, thereby facilitating understanding of the model dynamics.

Emergent properties of biological networks

This example[10] explores intracellular signaling of the MAP kinase pathway [18] to illustrate emergent behaviors such as hysteresis induced by covalent modifications of proteins, which serve as widespread regulatory mechanisms in living systems. T2B can facilitate understanding of the description of model components and demonstrate hysteresis by performing steady-state analysis for two different stable states. In this case study, depending on the initial parameters of two simulation runs, the system can reach two distinct steady states. This observation provides an intuitive, hands-on way to demonstrate complex system-level behaviors.

While T2B is a significant step in providing access to systems biology models, there are several limitations that would have to be addressed in the future releases. Although, T2B framework facilitates rapid reuse of ODE models by querying models on BioModels database or by directly uploading any ODE model in SBML format, it currently lacks functionality for expanding these models. The initial release of T2B is primarily designed for model screening and executing functional models.

Our benchmarking results revealed good Task Completion metrics score, averaging 0.7 across both sets of questions (Table S3). Here, a score of 0 represents failing the task, while a score of 1 indicates perfect accordance with the expected output. The slightly lower average reflects cases where minor variations in the phrasing of a prompt or representation of species' name affected the results. To achieve optimal performance, we recommend the users to adjust their prompts based on guidelines provided on our documentation page. For future work, we plan to expand the benchmarking suite to also evaluate tool and argument correctness for a more comprehensive evaluation of T2B's performance. We also plan to apply approaches commonly used to train AI agents, such as reinforcement learning, supervised fine-tuning, and LoRA, to improve T2B's task completion and tool-selection capabilities [7], [20].

Conclusions

T2B is an open-source, user-friendly agentic AI platform designed to facilitate access to computational models of biological systems. By providing a natural language interface, supporting open-source community standards, and integrating with BioModels data base, T2B reduces the barrier for non-experts to benefit from computational biology models and overall contributes to the FAIRification of models [26]. Furthermore, T2B supports computational modelers by simplifying testing and validation during

[9] Detailed prompts and their description are available at https://virtualpatientengine.github.io/AIAgents4Pharma/talk2biomodels/cases/Case_2/.

[10] Detailed prompts and their description are available at https://virtualpatientengine.github.io/AIAgents4Pharma/talk2biomodels/cases/Case_3/.

model development phase and encouraging model sharing in community-wide standardized format. In addition, when hosted on a web server, T2B also offers reviewers and researchers an installation-free way to evaluate models for correctness, validity, and reproducibility—areas that remain challenging in practice.

In this paper, we presented use cases in precision medicine, epidemiology, and the exploration of emergent system properties to demonstrate the ease at which T2B users can reproduce and explore mathematical modelling results.

We encourage contributions to our open-source project from the community.

Availability and requirements

Project name: Talk2Biomodels.

Project home page: https://github.com/VirtualPatientEngine/AIAgents4Pharma

Operating system(s): Platform independent.

Programming language: Python.

Other requirements: All the dependencies are listed in the project GitHub repository.

License: MIT.

Any restrictions to use by non-academics: not applicable.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12859-025-06310-1.

Supplementary Material 1.

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Author contributions

LW, GS and DM conceptualized the study. LW prototyped T2B functions. GS developed the agentic AI framework and the agentic AI tools for T2B. AWM contributed to the development of the agentic AI framework and created the figures and icons. RHS implemented ***get annotations*** tool. GS and AWM developed T2B's user interface. HH and TN crosslinked T2B in BioModels repository. DM, RSMS, and LW conceptualized use cases. LW designed use cases. LW wrote the manuscript and documentation of the use cases. DM supervised the writing of the manuscript and the development of the agentic AI framework. All authors contributed to testing T2B and reviewing the manuscript.

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Data availability and materials

The datasets analysed during the current study are available in the BioModels repository, https://www.ebi.ac.uk/biomodels/.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

MHM, HC, TA and TK are Sanofi employees and may hold shares and/or stock options in the company. No other conflicts of interest are declared.

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